Flight to Efficiency: Trajectory Optimization in SWIPT-Enabled UAV-Assisted NOMA for Future Wireless Networks

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***Abstract*—Integration of simultaneous wireless information and power transfer (SWIPT) enabled non-orthogonal multiple access (NOMA) in an unmanned aerial vehicle (UAV) is a novel variant that benefits enhanced spectrum efficiency and energy efficiency. Despite these potential advantages, resource allocations and UAV trajectory optimizations pose major challenges to its realization. Thus, in this work, we optimize the UAV's trajectory including its positioning, in up-link, and down-link NOMA-enabled UAV-assisted communication network while maximizing achievable rates at the ground users. To further enhance energy efficiency, the UAV is equipped with SWIPT to harvest energy from the received signals. The proposed solution utilizes the particle swarm optimization (PSO) tool to optimize the UAV's position to maximize the minimum achievable rate while maximizing SWIPT-based energy harvesting. Simulation results demonstrate the improved minimum achievable rate of the proposed solution making the proposed communication model effective over conventional UAV communications.**

***Index Terms*—Future Wireless Networks, NOMA, SWIPT, UAV-Assisted Communication**

I. Introduction

The rapid growth of wireless communication technologies and the rise of unmanned aerial vehicles (UAVs) have led to significant advancements in wireless networks [1]. It is evident that by combining simultaneous wireless information and power transfer (SWIPT) and non-orthogonal multiple access (NOMA) with UAV-assisted communications can improve overall performance in fifth-generation (5G) and beyond communication networks. UAV-assisted communication has attracted significant attention within the research community due to its unique advantages such as mobility, flexibility, cost-effectiveness, and aerial coverage in remote areas [1], [2]. UAVs can enhance network capacity and quality of service (QoS) by acting as flying Base Stations (BSs) or relays, utilizing existing communication infrastructure [3], [4]. They offer extensive coverage, enabling communication with remote, disconnected devices, and can be deployed on-demand to provide temporary communication services [5]–[7]. Integrating UAV capabilities with SWIPT and NOMA technologies paves the way for developing advanced, high-performance wireless communication networks.

SWIPT harnesses energy from radio frequency (RF) signals while maintaining QoS, making it valuable for UAV-assisted communications by replenishing batteries or powering communication capabilities, thus extending network operating periods [8]–[10]. NOMA improves wireless network connectivity, capacity, and spectrum efficiency by allowing multiple users to share resources simultaneously, enhancing spectrum utilization over traditional orthogonal multiple access (OMA) methods. Integrating NOMA with UAV-assisted communications increases system throughput, user experience, and network flexibility [11]–[13]. However, in SWIPT-enabled UAV-assisted NOMA networks, resource allocation and UAV trajectory optimization remain challenging.

Motivated by the above, this paper proposes a novel channel gain-based power allocation algorithm to enhance fairness in achievable rates for ground users (GUs) in downlink NOMA within SWIPT-enabled UAV-assisted communication networks. Then, we formulate a UAV trajectory optimization problem to maximize the minimum achievable rate, while maximizing the amount of energy harvested via SWIPT. Finally, we address the formulated optimization problem by employing particle swarm optimization (PSO) to determine the optimal UAV trajectory and hovering positions. Our results demonstrate the superior performance of the proposed solution compared to the predefined trajectory of the UAV in UAV-assisted communication networks, providing valuable insights into its effectiveness.

The remainder of this paper is organized as follows. Section II introduces the problem formulation of the proposed

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SWIPT-enabled UAV-assisted NOMA communication system and Section III shows the simulation results. Finally, Section IV summarizes this work with future directions.

II. System Model and Problem Formulation

As illustrated in Fig. 1, we consider a UAV-assisting wireless communication network between BSs and GUs utilizing uplink and downlink NOMA, where UAV is equipped with energy harvesting capabilities via SWIPT. UAV travels between its pre-defined starting and ending locations using the optimal trajectory generated, while maximizing minimum achievable GUs' downlink communication rates.

A. Preliminaries of the System Model

As in the proposed system model illustrated in Fig. 1, one UAV flying over N number of GUs while simultaneously establishing the downlink NOMA links with the GUs and the uplink NOMA links with the M number of BSs. The GUs in the proposed network is indexed by the set $\mathcal{N} = \{1, 2, \cdots, N\}$ and each of their locations can be expressed as $GU_n, \forall n \in \mathcal{N}$, $GU_n = [{}^{gu}x_n, {}^{gu}y_n] \in \mathbb{R}^{1\times 2}$, which ${}^{gu}x_n$ and ${}^{gu}y_n$ represent the position in horizontal plane of nth GU in the 3-dimention (3D) Cartesian coordinates. The locations of BSs are denoted as $BS_m, \forall m \in \mathcal{M}$, where $\mathcal{M} = \{1, 2, \cdots, M\}$. Horizontal and vertical coordinates of the mth BS are given by $BS_m = [{}^{bs}x_m, {}^{bs}y_m]$ and ${}^{bs}z_m$, respectively. In the simulation, the UAV follows a predefined trajectory with specified start and end positions. The UAV's position is denoted as $[Q, z_u]$ where $Q = [x_u, y_u]$ indicates the horizontal position of the UAV, and z_u represents the vertical position or height of the UAV from the ground level. As depicted in Fig. 1, the uplink involves data transmission from BSs to UAV, while the downlink involves data transmission from UAV to GUs. The UAV uses SWIPT in the uplink to harvest energy from the received signals from BSs supporting its energy requirement for downlink communications.

B. Channel Model

The proposed communication network involves two channels: the uplink between the mth BS and UAV is denoted by ${}^{bs}h_m$, and the downlink between the UAV and n^{th} GU is denoted by ${}^{gu}h_n$. The channel models for both uplink and downlink can be expressed as follows:

$$ {}^{gu}h_n = \sqrt{{}^{gu}\beta_n}\,{}^{gu}g_n, \tag{1a} $$

$$ {}^{bs}h_m = \sqrt{{}^{bs}\beta_m}\,{}^{bs}g_m, \tag{1b} $$

where ${}^{gu}\beta_n$ and ${}^{bs}\beta_m$ denote the large-scale average power gain in the downlink and uplink for both shadowing and path loss, respectively. These parameters account for signal attenuation, environmental factors, and the distance between transmitter and receiver providing a comprehensive understanding of how the environment and distance influence wireless signals [14]–[16]. The large-scale average power gain for the nth GU in the downlink channel and m^{th} BS in the uplink channel can be written as follows:

$$ {}^{gu}\beta_n = \beta_0\,{}^{gu}d_n^{-\zeta}, \tag{2a} $$

$$ {}^{bs}\beta_m = \beta_0\,{}^{bs}d_m^{-\zeta}, \tag{2b} $$

where β_0 denotes the average power gain at a reference distance of $d_0 = 1m$, serving as a baseline for comparing power gain at other distances. The path loss exponent ζ determines how quickly the received signal weakens with increasing distance, typically ranging from 2 to 6 depending on propagation conditions. The line-of-sight (LoS) distances between the UAV and the nth GU, and the mth BS are given by ${}^{gu}d_n$ and ${}^{bs}d_m$, respectively, and can be further expressed as ${}^{gu}d_n = \sqrt{|Q - GU_n|^2 + {z_u}^2}$ and ${}^{bs}d_m = \sqrt{|Q - BS_m|^2 + |z_u - {}^{bs}z_m|^2}$.

In addition to signal attenuation, multi-path propagation causes signal amplitude and phase changes, resulting in rapid fluctuations in received signal strength. These fluctuations are represented by small-scale fading coefficients ${}^{gu}g_n$ for n^{th} GU in the downlink channel and ${}^{bs}g_m$ for m^{th} BS in the uplink channel. In this work, we use the Rician fading model to accurately represent small-scale fading, accounting for both LoS and non-LoS channels. This model considers direct and scattered signals, making it ideal for simulating signal propagation between UAV to GUs, and BSs to UAV. The Rician fading channels for UAV to n^{th} GU and m^{th} BS to UAV can be modeled as follows:

$$ {}^{gu}g_n = \sqrt{\frac{{}^{gu}K_n}{{}^{gu}K_n + 1}}\,g + \sqrt{\frac{{}^{gu}K_n}{{}^{gu}K_n + 1}}\,\tilde{g}, \tag{3a} $$

$$ {}^{bs}g_m = \sqrt{\frac{{}^{bs}K_m}{{}^{bs}K_m + 1}}\,g + \sqrt{\frac{{}^{bs}K_m}{{}^{bs}K_m + 1}}\,\tilde{g}, \tag{3b} $$

where the channel between the UAV and n^{th} GU and the channel between m^{th} BS and the UAV are characterized by two components. The first component is deterministic and represents the LoS signal denoted as g. The second component is random and accounts for the NLoS signals, represented by $\tilde{g}$. In particular, the scattered component $\tilde{g}$ follows a circularly symmetric complex Gaussian (CSCG) distribution with zero mean and unit variance. This random variable captures the variability and unpredictability of NLoS signals, which are influenced by environmental factors, obstacles, and signal reflections. The Rician factor, denoted as ${}^{gu}K_n$ and ${}^{bs}K_m$ for n^{th} GU in downlink and m^{th} BS in uplink channels respectively. The Rician factors can vary in the different positions of the UAV. In general, increasing the elevation angle results in a bigger part of the LoS component and a higher Rician factor [17]. The angle-dependent Rician factor can be estimated using an exponential function as given in [17], [18]. The angle-dependent Rician factor can be modeled using the exponential functions, ${}^{gu}K_n = A_1 e^{A_2\,{}^{gu}\theta_n}$ and ${}^{bs}K_m = A_1 e^{A_2\,{}^{bs}\theta_m}$, where ${}^{gu}\theta_n = \sin^{-1}\left(\frac{z_u}{{}^{gu}d_n}\right)$ and ${}^{bs}\theta_m = \sin^{-1}\left(\frac{z_u - {}^{bs}z_m}{{}^{gu}d_n}\right)$, which represent the elevation angle for the nth GU and m^{th} BS with the UAV position respectively. The Rician factor is denoted as ${}^{gu}K_n$ and ${}^{bs}K_m$, and is constrained within specific bounds determined by coefficients A_1 and A_2. Precisely, the Rician factor lies within the range $K_{min} \leq [{}^{gu}K_n, {}^{bs}K_m] \leq$

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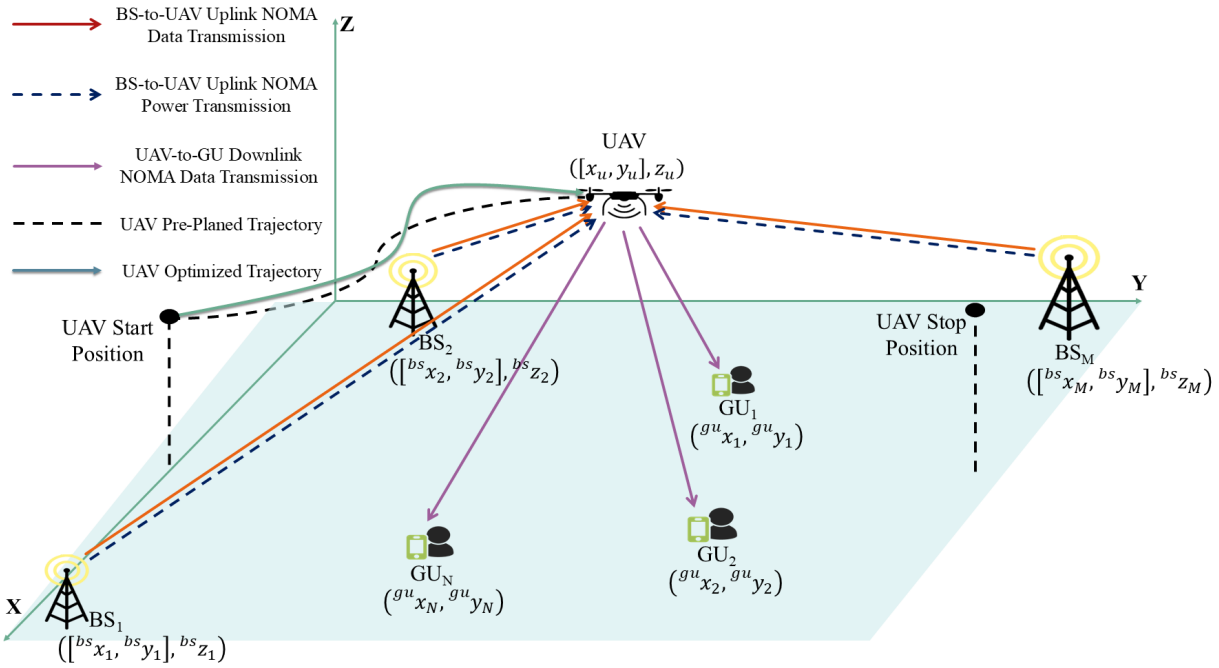


Fig. 1: An illustration of the proposed NOMA and SWIPT enabled UAV-assisted communication network.

K_{max}, where K_{min} is the minimum Rician factor and K_{max} is the maximum Rician factor. These constant factors A_1 and A_2 are environment-specific and can be computed based on channel characteristics and experimental setup as in $K_{min} = A_1$ and $K_{max} = A_1 e^{A_2 \frac{\pi}{2}}$ [18].

C. Energy Harvesting Model

This section provides the energy harvesting model of the UAV's, focusing on extracting energy from BS signals using power-splitting methods in SWIPT. The harvested energy is then used for downlink data transmission. The amount of energy harvested by the UAV from the received signals can be written as $E_H = \eta\varphi P_r T_{Uplink}$, where T_{Uplink} is uplink data transmission time and P_r is received total power at the UAV from the BSs. The total received power at the UAV can be written as $P_r = P_{BS}\left(\sum_{m=1}^{M} \left|^{bs}h_m\right|^2\right)$, where P_{BS} is the transmit power at each BS. Considering the energy harvested at the UAV from uplink signals, the transmit power of the UAV can be written as $P_t = \frac{E_H}{T_{Downlink}}$ where $T_{Downlink}$ denotes the data transmission time of the downlink communication[1]. Therefore, the transmit power of the downlink channel in the energy harvesting model can be rewritten as $P_t = \eta\varphi P_r$.

D. Data Collection Model

This section outlines a data transmission process consisting of two phases: from BSs to UAV (uplink NOMA principles) and from UAV to GUs (downlink NOMA principles).

[1]For simplicity, we have considered equal time fractions for both downlink and uplink communication

1) Uplink NOMA: In uplink NOMA, the BSs transmit signals to the UAV with transmit power P_{BS}. The UAV uses successive interference cancellation (SIC) to decode the superimposed signals, and the achievable rate of each BS is calculated based on the received signal's power level at the UAV. Accounting for signal reception strength from individual BSs and the interference from other BSs, the achievable rate at UAV for signal received from m^{th} BS in bits per second (bps) within this system model can be expressed as follows:

$$^{bs}R_m = \log_2\left(1 + \frac{P_{BS}\left|^{bs}h_m\right|^2}{P_{BS}\left(\sum_{k=m+1}^{M}\left|^{bs}h_k\right|^2\right) + \sigma^2}\right), \quad (4)$$

where σ^2 is additive white Gaussian noise (AWGN) power at the receiver. Here, we assumed that the numbering of each BS is in ascending order based on the strength of their respective channel conditions.

2) Downlink NOMA: In downlink NOMA, the UAV merges signals from each BS into a superimposed signal, assigning power coefficients based on the channel conditions. Power control strategies maximize achievable rates while preserving interference control and achieving QoS standards. Upon receiving the superimposed signal, GUs decode their signals using the SIC process, eliminating signals with higher power coefficients treating lower coefficients as interference. Each GU is numbered in descending order based on channel strength. After applying the SIC process at each GU, the achievable rate for n^{th} GU in bps within this system model

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can be mathematically expressed as follows:

$$ {}^{gu}R_n = \log_2 \left(1 + \frac{P_{GU} \alpha_n |{}^{gu}h_n|^2}{P_{GU} \left(\sum_{k=1}^{n-1} \alpha_k \right) |{}^{gu}h_n|^2 + \sigma^2} \right), \quad (5) $$

where P_{GU} refers to the transmit signal power from the UAV and α_n represents the allocated power coefficient for n^{th} GU in the downlink NOMA network.

Algorithm 1 Algorithm for allocating power coefficients via fractional value

$i \leftarrow 1$
$\delta \leftarrow 0.8$
${}_{sorted}h \leftarrow sort(h)$
while $i \leq n$ **do**
  $index \leftarrow find(h ==_{sorted} h_i)$
  if $n == 1$ **then**
    $coef_{index} \leftarrow \delta$
  else if $i < n$ **then**
    $coef_{index} \leftarrow \delta(1 - \sum_{j=1}^{n} coef_j)$
  else
    $coef_{index} \leftarrow (1 - \sum_{j=1}^{n} coef_j)$
  end if
  $i \leftarrow i + 1$
end while

E. Problem Formulation

The main goal is to increase the achievable rate for GUs on the NOMA downlink wireless network, using a system model to determine the optimal UAV trajectory. For ease of comprehension in this study, the system model was constructed within a virtual environment measuring W in width, L in length, and H in height in a 3D Cartesian plane. The BSs and GUs are positioned within this 3D Cartesian plane in accordance with the previously specified location vectors for both BSs and GUs. For this study, it is considered that there are three BSs, three GUs, and a single UAV. At the outset of the simulation, the UAV's path was pre-established, with defined starting and ending positions: the UAV's starting position is given as $([x_{u_0}, y_{u_0}], z_{u_0})$, and its ending position is $(([x_{u_l}, y_{u_l}], z_{u_l})$ in the 3D Cartesian plane. Throughout the straight journey of the UAV from the UAV start position to the UAV end position, the system requirement mandates that the UAV advances by $y_{i_{max}}$ horizontally towards the UAV end position with each step i. The UAV's vertical movement is limited, ranging between a minimum height of $z_{u_{min}}$ and a maximum height of$z_{u_{max}}$.

F. Power Coefficients for GUs

In each simulation step, the UAV allocates power coefficients to GUs based on their LoS distances and channel conditions using CSI. The study assessed performance with methods detailed in Algorithm 1 and Algorithm 2. Algorithm 1 sorts channel gains and assigns power coefficients using a fractional value. Algorithm 2 calculates power coefficients by first normalizing based on channel gains and then normalizing these coefficients to ensure the sum of power coefficients equals 1.

Algorithm 2 Algorithm for allocating power coefficients using channel gain

$i \leftarrow 1$
while $i \leq n$ **do**
  $coef_i \leftarrow 1 - \frac{h_i}{\sum_{k=1}^{n} h}$
  $i \leftarrow i + 1$
end while
$j \leftarrow 1$
while $j \leq n$ **do**
  $coef_j \leftarrow 1 - \frac{coef_j}{\sum_{k=1}^{n} coef_k}$
  $j \leftarrow j + 1$
end while

G. UAV Positioning

The UAV is positioned in the 3D Cartesian plane to maximize the minimum achievable rate for the downlink NOMA system, determined using an energy harvesting model, and the objective function for this is expressed as follows:

$$ \begin{aligned} \underset{x_u, y_u, z_u}{\text{maximize}} \quad & f\left(\min\left({}^{gu}R_1, {}^{gu}R_2, \cdots, {}^{gu}R_N\right)\right) && (6a) \\ \text{subject to} \quad & x_{u_i} - x_{u_{i-1}} \leq x_{i_{max}}, && (6b) \\ & y_{u_i} - y_{u_{i-1}} = y_{i_{max}}, && (6c) \\ & z_{u_{min}} \leq z_{u_i} \leq z_{u_{max}}, && (6d) \\ & 1 \leq i \leq \frac{L}{y_{i_{max}}} + 1 && (6e) \end{aligned} $$

where x_{u_i} , y_{u_i} and z_{u_i} are the UAV position in x , y and z plane at i^{th} step. As per each constraint, the UAV's vertical position should be between $z_{u_{min}}$ to $z_{u_{max}}$ and the UAV should fly $y_{i_{max}}$ in y direction (along the y plane) in every step, and the displacement of x plane should be less than or equals to $x_{i_{max}}$. Similarly, the objective function for finding the optimum position for the UAV to maximize the minimum achievable rate in downlink NOMA communication with energy harvesting can be expressed as:

$$ \begin{aligned} \underset{x_u, y_u, z_u}{\text{maximize}} \quad & f\left(\min\left({}^{gu}R_{1e}, {}^{gu}R_{2e}, \cdots, {}^{gu}R_{Ne}\right)\right) && (7a) \\ \text{subject to} \quad & x_{u_i} - x_{u_{i-1}} \leq x_{i_{max}}, && (7b) \\ & y_{u_i} - y_{u_{i-1}} = y_{i_{max}}, && (7c) \\ & z_{u_{min}} \leq z_{u_i} \leq z_{u_{max}}, && (7d) \\ & 1 \leq i \leq \frac{L}{y_{i_{max}}} + 1 && (7e) \end{aligned} $$

Here, ${}^{gu}R_{ne}$ is the achievable rate of nth GU in with the energy harvesting model which can be represented as:

$$ {}^{gu}R_{en} = \log_2 \left(1 + \frac{P_{GUe}\ \alpha_n\ |{}^{gu}h_n|^2}{P_{GUe} \left(\sum_{k=1}^{n-1} \alpha_k \right) |{}^{gu}h_n|^2 + \sigma^2} \right), \quad (8) $$

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TABLE I: Simulation parameters used in the simulation setup of the proposed communication network.

Parameter Name	Value
System model environment width (W)	$2000m$
System model environment length (L)	$2000m$
System model environment height (H)	$320m$
Number of GUs (N)	3
Number of BSs (M)	3
Number of UAVs	1
Path loss exponent (ζ)	2.5
Reference channel gain (β_0)	$-50dB$
Minimum Rician factor (K_{min})	$4dB$
Maximum Rician factor (K_{max})	$12dB$
Energy harvesting efficiency (η)	0.7
Power splitting coefficient in SWIPT (φ)	0.7
BS's transmit power (P_{BS})	$46dBm$
UAV's transmit power (P_{GU})	$20dBm$
Initial power coefficient factor (δ)	0.8
UAV start position ($([x_{u_0}, y_{u_0}], z_{u_0})$)	$([1000, 2000], 200)$
UAV end position ($([x_{u_l}, y_{u_l}], z_{u_l})$)	$([1000, 0], 200)$
UAV minimum height ($z_{u_{min}}$)	$100m$
UAV maximum height ($z_{u_{max}}$)	$300m$
UAV displacement in x plane ($x_{i_{max}}$)	$500m$
UAV displacement in y plane ($y_{i_{max}}$)	$100m$

where $P_{GU\,e}$ is transmitted total power obtained from the harvested energy by the UAV and it can be represented as $P_{GU\,e} = \eta\varphi P_{BS}\left(\sum_{m=1}^{M}\left|{}^{bs}h_m\right|^2\right)$.

Given the non-convex nature of the problem, particle swarm optimization (PSO) [19] is used to optimize UAV trajectories. PSO efficiently explores the solution space, iteratively updating particle positions and velocities to maximize the minimum achievable rate in the downlink NOMA network.

III. Simulation Results

In this section, we present simulation results evaluating the integration of SWIPT-enabled UAV-assisted NOMA in future wireless networks, aiming to maximize the minimum achievable rate for GUs in downlink NOMA communication. The results demonstrate that our proposed solution effectively improves the minimum achievable rate for GUs. All simulation parameters not otherwise specified are listed in Table I. In this work, we assume the UAV has enough battery life for its trajectory and communication tasks, and it uses SWIPT technology to harvest energy from BSs for transmitting signals to GUs.

First, the performance of the achievable rates at the GUs in downlink NOMA with respect to the transmit power of the BS are plotted under different power coefficient allocations methods as given in Fig. 2. The proposed channel gain-based power coefficient allocation method decides on the power coefficient based on the Rician fading channel gains between UAV and the GU. This method follows the proposed Algorithm 2 to allocate the power coefficients for each GU in downlink NOMA communication. The fractional power coefficient method allocates the power coefficients at the UAV in downlink NOMA utilizing Algorithm 1, while the fixed power allocation method uses predefined parameters to allocate power coefficients for GUs in downlink NOMA

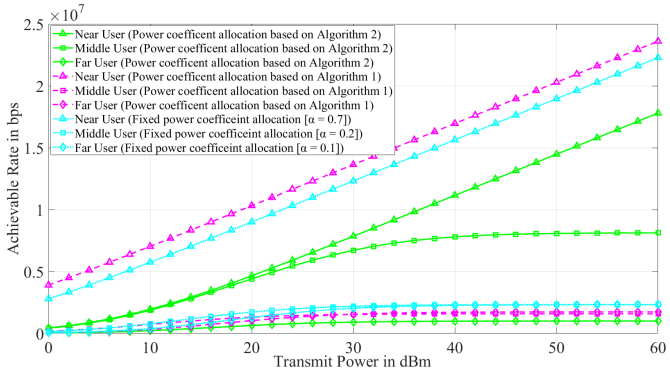


Fig. 2: Achievable rate at the GUs for different power coefficient allocation methods versus transmit power.

communication. As can be seen in Fig. 2, the channel gain-based power coefficient allocation method provides higher achievable rates for two users out of three users, while the other two methods achieved higher achievable rates for only one user (near user). Hence, this method shows fair performance compared to the other two methods. Thus, here onwards channel gain-based power coefficient allocation method is used in the rest of the simulations.

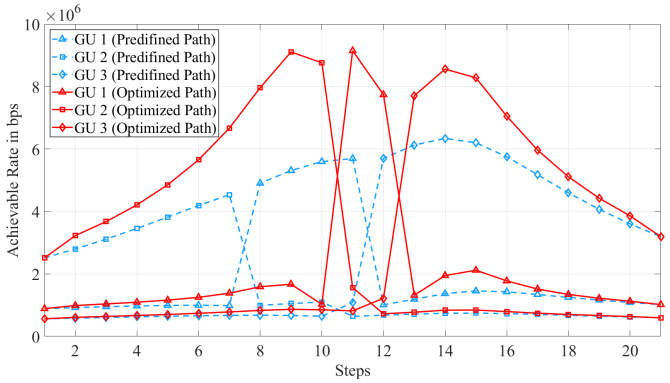


Fig. 3: Achievable rate for each GU through predefined and optimized UAV's trajectory without SWIPT-enabled model

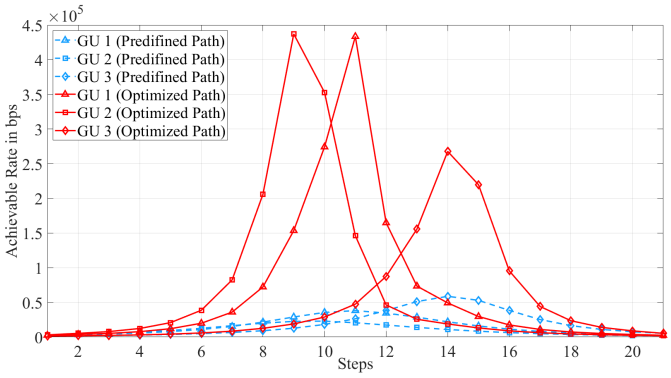


Fig. 4: Achievable rate for each GU through predefined and optimized UAV's trajectory with SWIPT-enabled model

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The results given in Fig. 3 and Fig. 4 compare the achievable rates at the each GU along the predefined and the optimized trajectory of the UAV with and without SWIPT, respectively. Regardless of enabling SWIPT, it can be observed from the both figures that when UAV follows the proposed optimized path generated via PSO always achieves higher rates at the GU compared to the predefined trajectory. Sudden increases and drops in rate illustrated in the figures occur due to the rapid changes of power allocation in the NOMA network throughout the UAV's optimized path. This occur due to changes in the channel conditions and LoS distance between GU and the UAV when serving each GU. Regardless of these rate changes, UAV travelling in the proposed optimized trajectory always shows superior rate performance compared to the conventional pre-defined trajectory. The achievable rates with SWIPT are illustrated in the Fig. 4. It can be seen from the figure that the achievable rate for each GU in the downlink NOMA communication with SWIPT has achieved higher rates within the optimal trajectory of the UAV compared to predefined trajectory. However, due to the power trade-offs in SWIPT enabled system, overall achievable rates given in Fig. 4 are lower compared to the achievable rates in Fig. 3. The reason is that the SWIPT-enabled model allocates total transmit power for GUs' superimposed signals using harvested energy from the BSs' signals, reducing the total transmit power allocation to GUs. This results in a decline in achievable rates for GUs within the NOMA downlink network. However, it is noteworthy that regardless of the decline in achievable rate in the SWIPT-enabled system, our proposed UAV trajectory optimization solution maintains required minimum achievable threshold at GU showing superior performance to predefined trajectory.

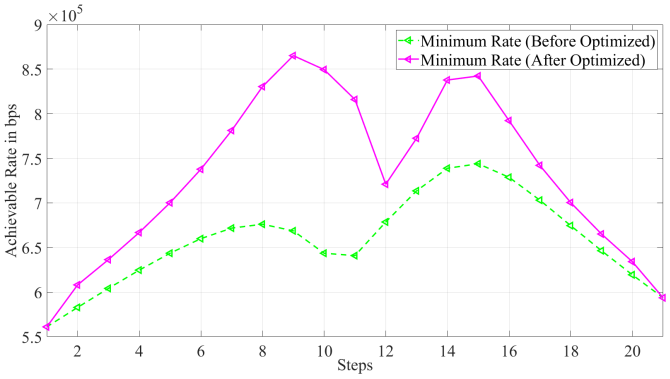


Fig. 5: Minimum achievable rates of the downlink NOMA network at the GUs in each step through predefined and optimized UAV's trajectory without SWIPT

Minimum achievable rates of the downlink NOMA network at the GUs via predefined and optimized UAV trajectory with and without SWIPT are illustrated in Fig. 5 and Fig. 6, respectively. It can be seen from both figures that UAVs traveling in an optimized trajectory always achieve superior minimum achievable rates compared to the UAV traveling in the predefined trajectory regardless of the presence of SWIPT.

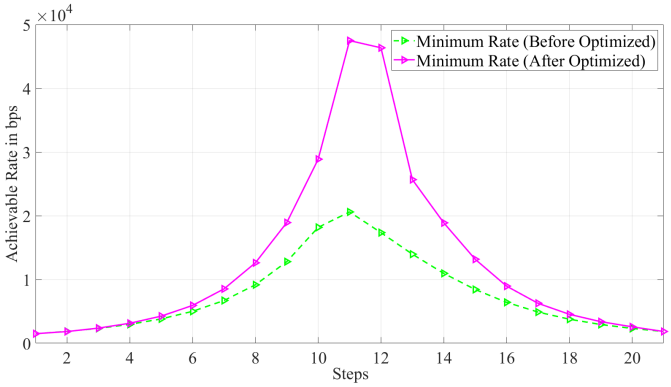


Fig. 6: Minimum achievable rates of the downlink NOMA network at the GUs in each step through predefined and optimized UAV's trajectory with SWIPT

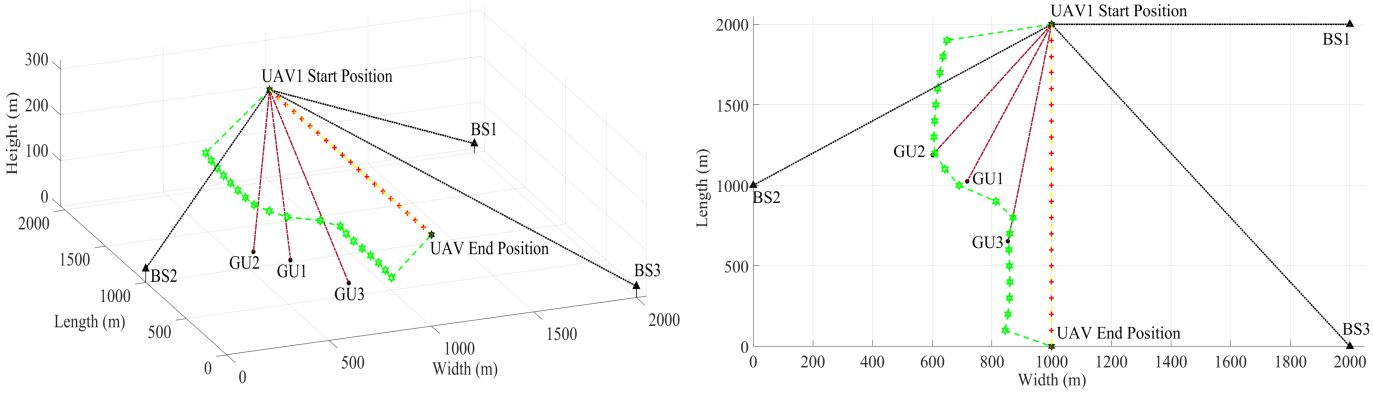
This validates the performance superiority of the proposed solution compared to conventional UAV-assisted communications.

Finally, for better understanding, 3D and bird's-eye views of both predefined and optimized UAV trajectories are given in Fig. 7(a) and Fig. 7(b), respectively. The UAV's predefined path is represented by yellow dashed lines, with red marks for each point on the straight trajectory. Green dashed lines with green star marks show the optimized path. Black dots and triangles represent GUs and BSs respectively. It can be observed from both figures due to the random nature of the channel conditions, UAVs favor to flying following a trajectory closer to GU while maintaining a minimum LoS distance to GU to maximize the minimum achievable rate. This can be easily understood by referring to Algorithm 1 and (7a).

IV. Conclusion and Future Directions

In this paper, we explored the integration of SWIPT-enabled UAV-assisted downlink and uplink NOMA, highlighting its potential to improve network performance, energy efficiency, and connectivity. The proposed system consisted of a UAV, three GUs, and three BSs, where the UAV acts as a relay to assist communication between BSs and GUs employing uplink and downlink NOMA together with SWIPT. First, we proposed a new algorithm based on channel gain-based power coefficient allocation for downlink NOMA at the UAV. Then, we optimized the UAV trajectory of the proposed communication setup using particle swarm optimization to improve the minimum achievable rate at the GUs. The simulation results proved the performance superiority of the proposed algorithm and the optimized UAV trajectory compared to the predefined UAV trajectory. This work can be further extended by incorporating multiple UAVs, GUs, and BSs into the proposed setup, while jointly optimizing the trajectories of the UAVs alongside energy efficiency to achieve the maximum minimum achievable rates at the GUs. This extension would better reflect practical applications of SWIPT in UAV communications utilizing NOMA, enhancing spectrum efficiency and effective use of radio resources.

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(a) 3D view of the UAV trajectory

(b) Bird's eye view of the UAV trajectory

Fig. 7: Optimal trajectory of the UAV versus predefined trajectory of the UAV.

Future research could focus on addressing challenges such as reducing the increased computational complexity of optimizing multiple UAV trajectories, ensuring robust communication links in dynamic environments, and managing interference between UAVs. Practical applications include disaster recovery operations where multiple UAVs provide network coverage, smart agriculture with UAVs monitoring and communicating with numerous ground sensors, and urban traffic management where UAVs coordinate with ground units to optimize traffic flow. Additionally, integrating advanced machine learning techniques to adaptively optimize network performance in real-time scenarios, and exploring hybrid energy sources for UAVs to extend their operational time could provide significant improvements.

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